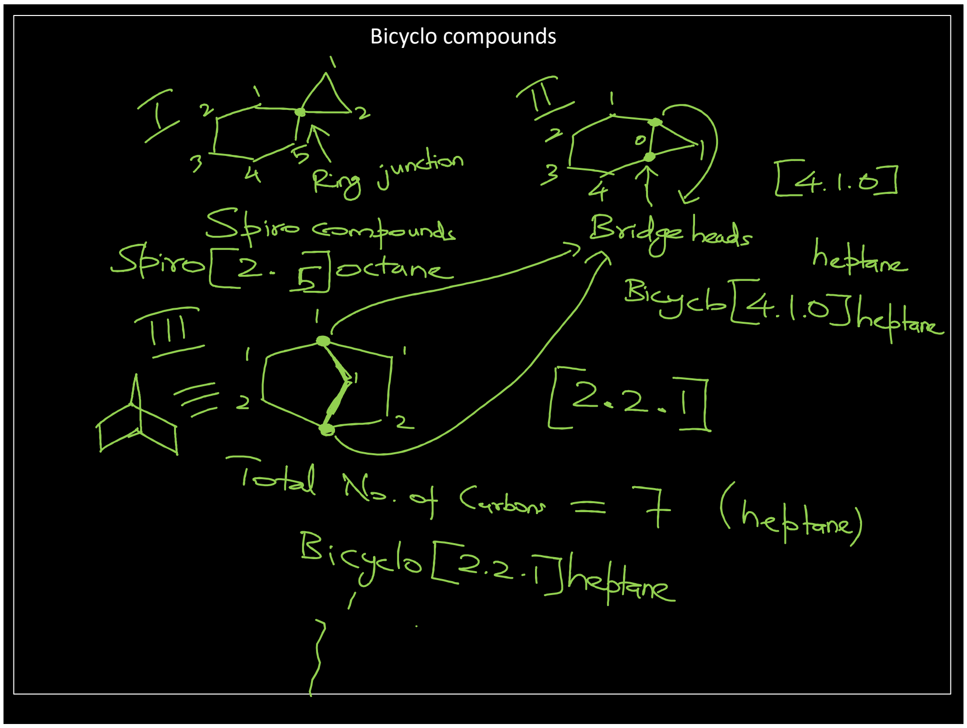
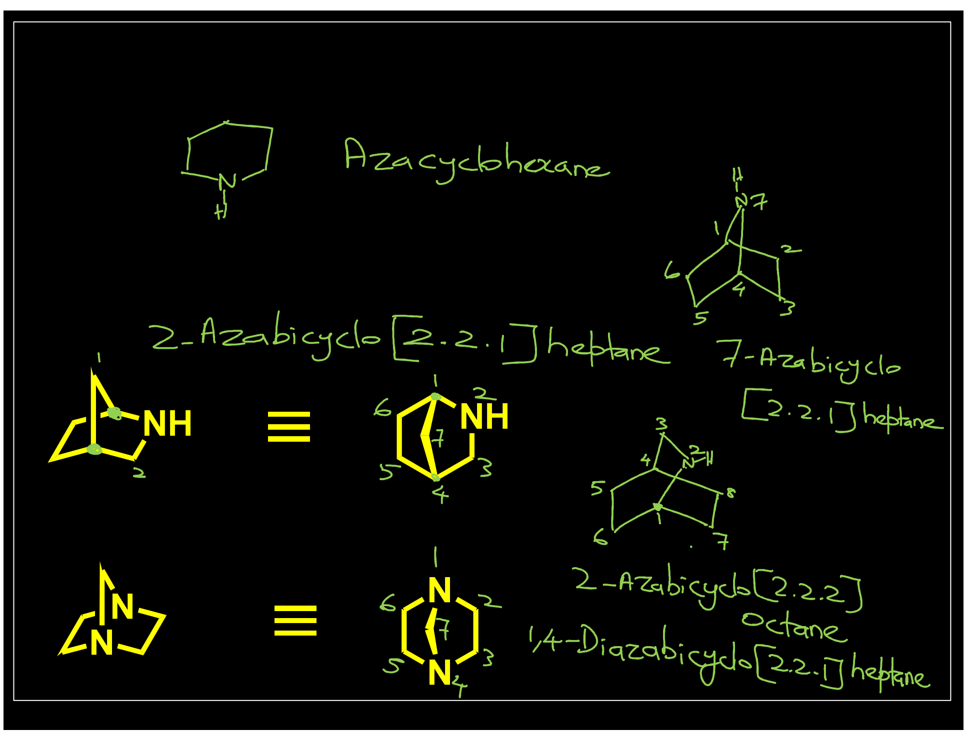
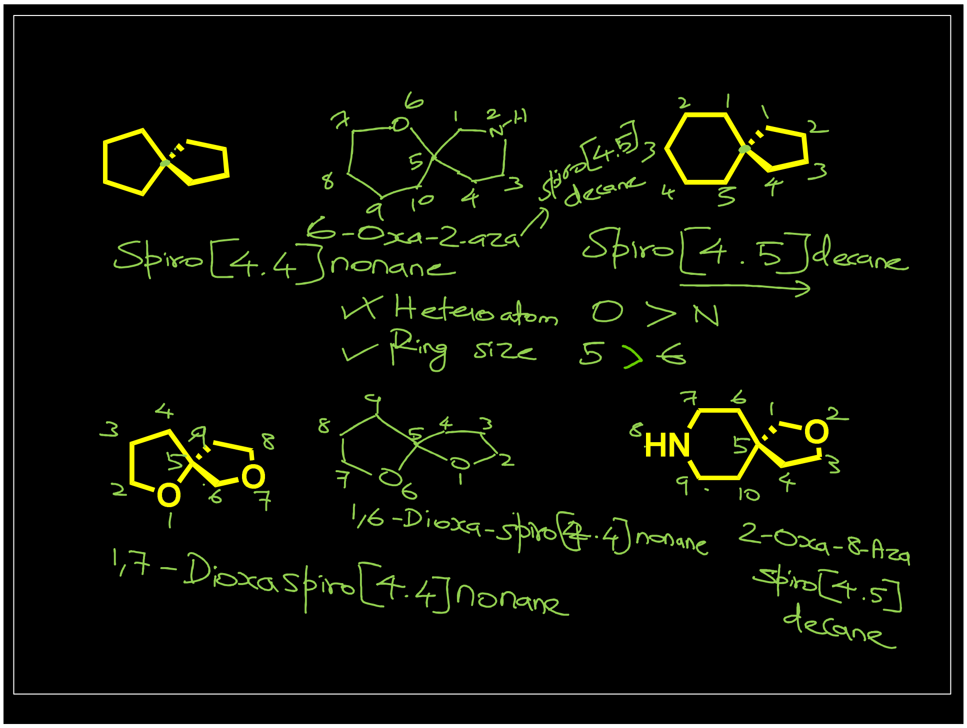
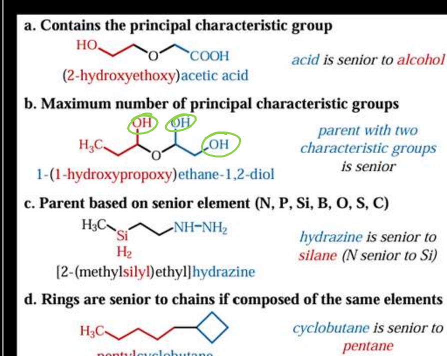
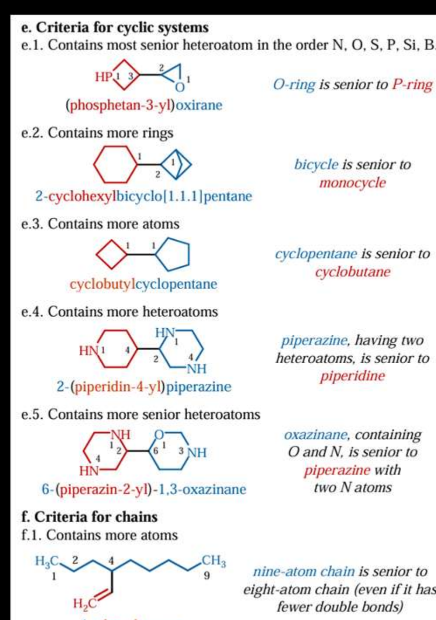

Bicyclo compounds
I
Ring junction
Spiro compounds
Spiro[2.5]octane
II
Bridge heads
[4.1.0]
heptane
Bicyclo[4.1.0]heptane
III
[2.2.1]
Total No. of Carbons = 7 (heptane)
Bicyclo[2.2.1]heptane


Azacyclohexane
2-Azabicyclo[2.2.1]heptane
7-Azabicyclo [2.2.1]heptane
NH
2-Azabicyclo[2.2.2] octane
1,4-Diazabicyclo[2.2.1]heptane
N
N

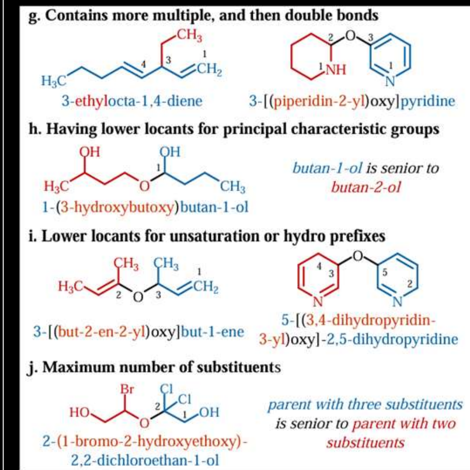


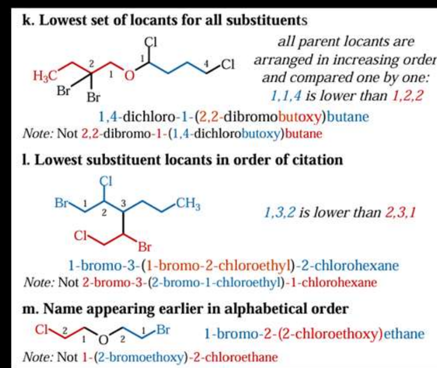
Seniority (higher preference) of parent compounds





Seniority (higher preference) of parent compounds





Seniority (higher preference) of parent compounds

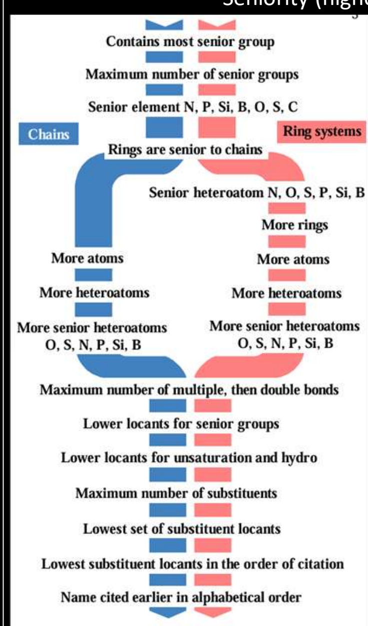
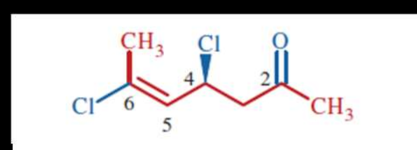


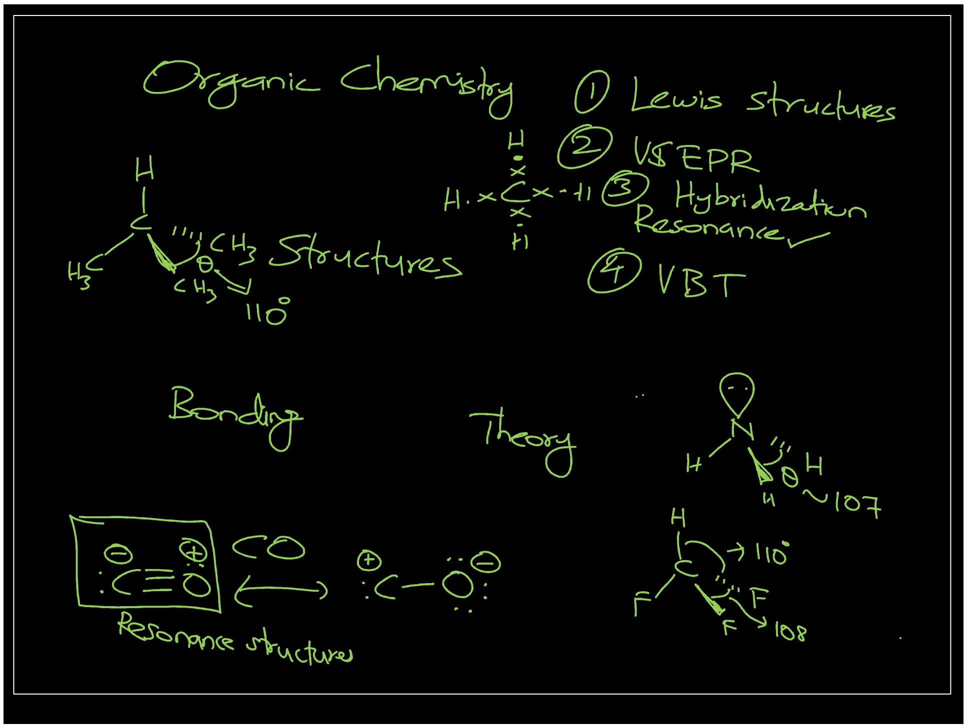
Figure 1: Criteria for choosing the senior parent compound



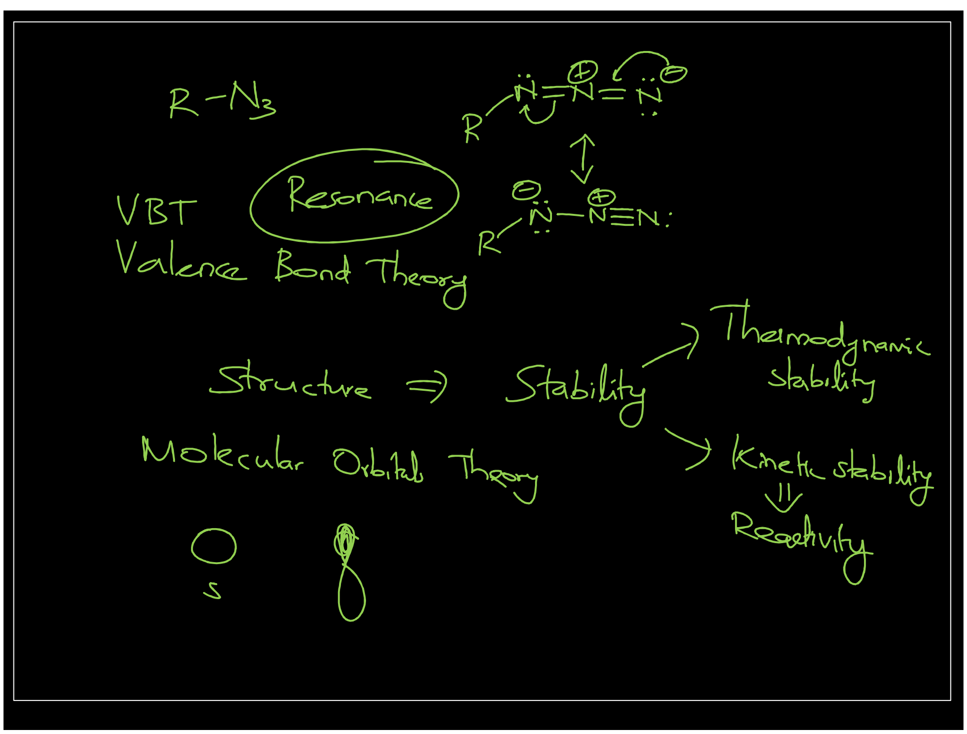
(4*S*,5*E*)-4,6-dichlorohept-5-en-2-one

hept(a)	parent (heptane)
en(e)	unsaturation ending
di	multiplicative prefix
2 4 5 6	locants
one	suffix for principal characteristic group
chloro	substituent prefix
E S	stereodescriptors
( )	enclosing marks

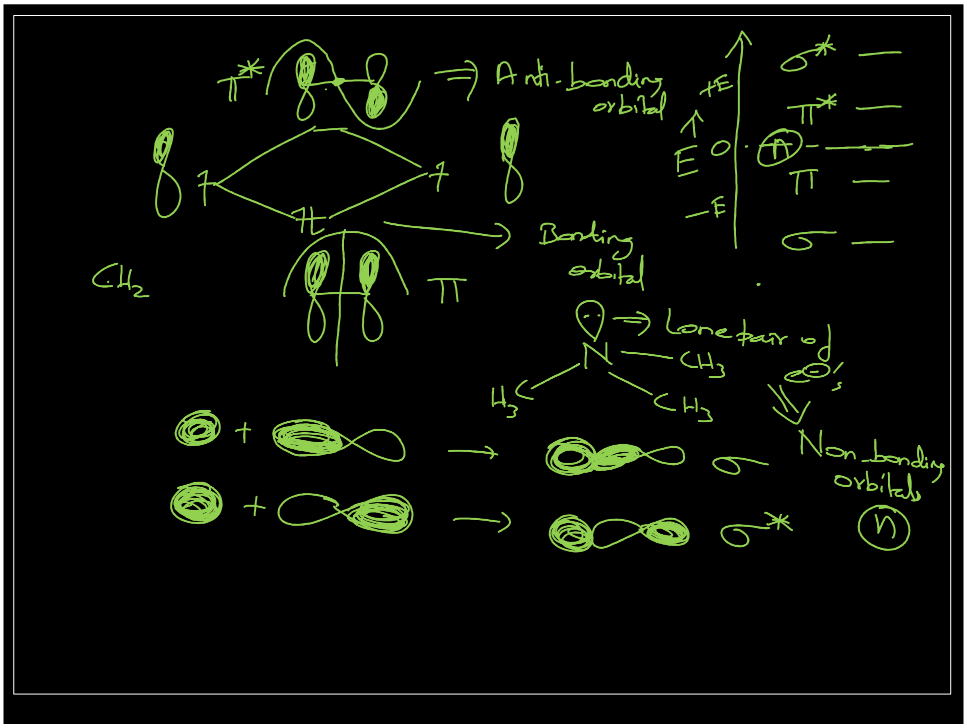
Pure Appl. Chem. **2020**; *92*(3): 527–539

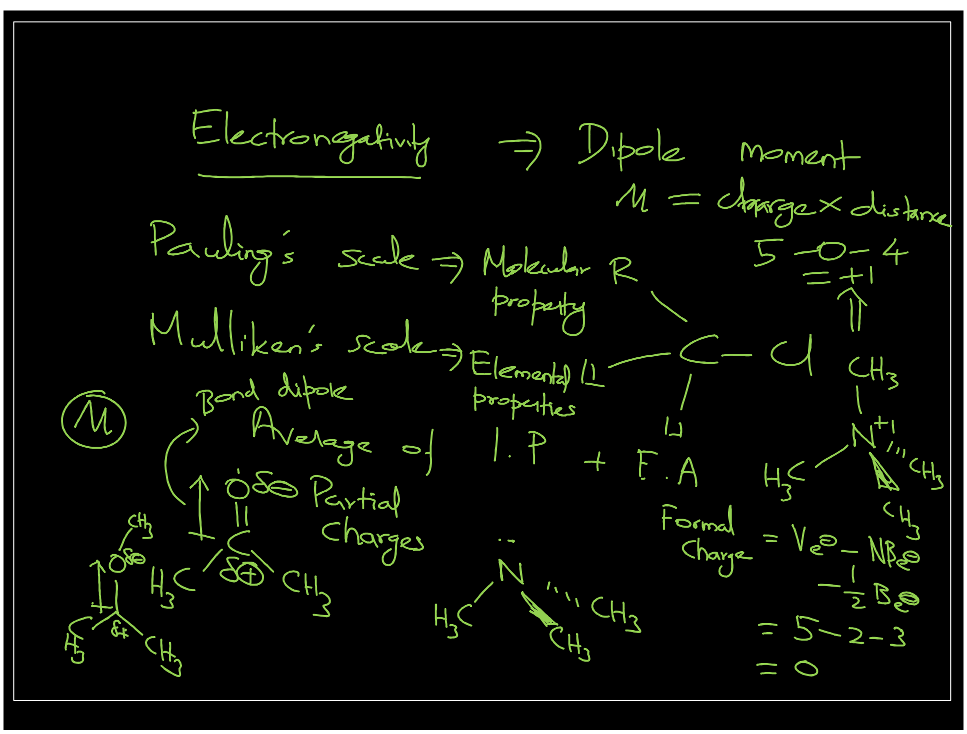


Organic Chemistry
① Lewis structures
② VSEPR
③ Hybridization
Resonance
④ VBT
Structures
110°
Bonding
Theory
~107
110°
108
CO
Resonance structures



R−N3
VBT
Resonance
Valence Bond Theory
Structure ⇒ Stability
Thermodynamic stability
Kinetic stability
Reactivity
Molecular Orbital Theory
s


π*
Anti-bonding orbital
π
Bonding orbital
CH2
+E
E
0
-E
σ*
π*
n
π
σ
Lone pair of e⁻'s
N
CH3
H3C
CH3
Non-bonding orbitals
n
σ
σ*


Electronegativity ⇒ Dipole moment
μ = charge × distance
Pauling's scale ⇒ Molecular property
Mulliken's scale ⇒ Elemental properties
Average of I.P + E.A
Bond dipole
μ
δ⊖ Partial Charges
δ⊕
CH3
H3C
R
C — Cl
5 − 0 − 4
= +1
N+1
Formal Charge = Ve⊖ − NBe⊖ − ½ Be⊖
= 5 − 2 − 3
= 0

Concepts involving acidity, basicity, tautomerism, inductive, mesomeric, Nathan field and steric effects

Electronegativity:

- Useful to predict the charge distribution in an organic molecule.
- Linus Pauling original concept: "*the power of an atom in a molecule to attract electrons to itself*"
- Pauling assigned values to various atom types based on bond dissociation energies of molecules.
- Pauling electronegativity scale depends upon *molecular* properties
- Mulliken's electronegativity scale: "derived from the average of the ionization potential (*the energy required to remove an electron from an atom or molecule*) and electron affinity (*the amount of energy released or required to attach another electron to an atom or molecule*) of an atom"
- Mulliken's electronegativity is an *atomic* property.

Electronegativities of Atoms According to the Scales of Pauling and Mulliken*

Atom	Pauling	Mulliken
H	2.1	3.01
B	2.0	1.83
C	2.5	2.67
N	3.0	3.08
O	3.5	3.22
F	4.0	4.44
Cl	3.0	3.54
Br	2.8	3.24
I	2.5	2.88
Li	1.0	1.28
Na	0.9	1.21
K	0.8	1.03
Mg	1.2	1.63
Ca	1.0	1.30
Al	1.5	1.37
Si	1.8	2.03
P	2.1	2.39
S	2.5	2.65

(Polar) C−F > C−Cl

C−H (0.4)

Polarization

Group electronegativity

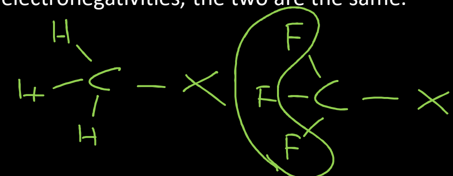
Table 1.2
Group Electronegativities, Scaled to be Compatible with the Pauling Scale*

Group	Electronegativity
CH_3 ✓	2.3
CH_2Cl ✓	2.8
$CHCl_2$ ✓	3.0
CCl_3 ✓	3.0
CF_3	3.4
Ph	3.0
$CH{=}CH_2$	3.0
$C{\equiv}CH$	3.3
$C{\equiv}N$	3.3
NH_2	3.4
NH_3^+	3.8
NO_2	3.4
OH	3.7

*Wells, P. R. "Group Electronegativities." *Prog. Phys. Org. Chem.*, 6, 111 (1968).

Some group or particular portions of a molecule have their own electronegativity. Example, CF_3 group vs CH_3 group

CF_3 group affect the charge distribution in a molecule via induction much more than a CH_3 group but if we consider only carbon electronegativities, the two are the same.

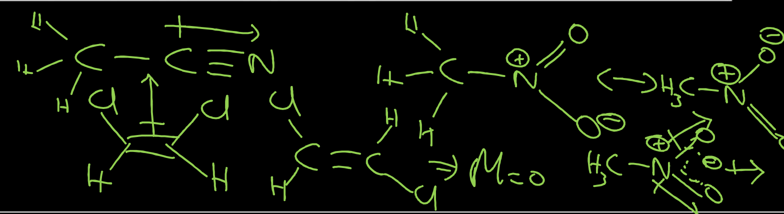


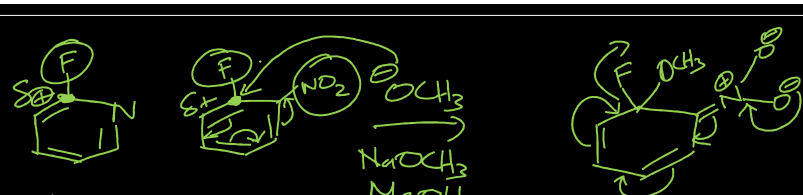
$\mu = q \times r$

Table 1.3
Molecular Dipole Values*

Compound	Molecular dipole (D)	Compound	Molecular dipole (D)
CCl_4	0.0	CH_3COCH_3	2.9
$CHCl_3$	1.0	CH_3COOH	1.7
CH_2Cl_2	1.6	CH_3COCl	2.7
CH_3Cl	1.9	CH_3COOCH_3	1.7
CH_3F	1.8	C_6H_5Cl	1.8
CH_3Br	1.8	$C_6H_5NO_2$	4.0
CH_3I	1.6	1-Butene	0.34
CH_3OH	1.7	1-Propyne	0.80
CH_3OCH_3	1.3	*cis*-2-Butene	0.25
CH_3CN	4.0	*cis*-1,2-Dichloroethene	1.9
CH_3NO_2	3.4	Tetrahydrofuran	1.6
CH_3NH_2	1.3	Water	1.8

Handbook of Chemistry and Physics, CRC Press, Inc., Boca Raton, FL (1979).





Nucleophile
$e^{\ominus}$ rich species

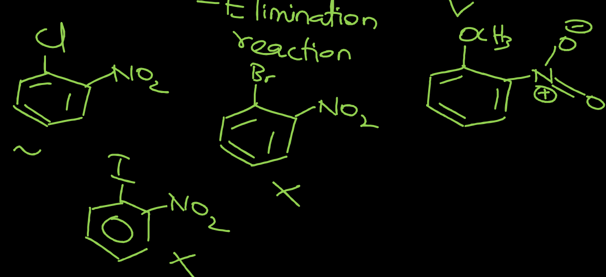
S_N^{Ar}

Adduct
Addition product

$- F^{\ominus}$

S_N1
S_N2
S_Ni

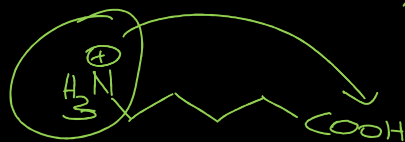
Addition
– Elimination
reaction



Inductive and Field effect

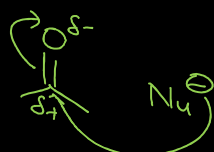
C → F ⇒ σ – $e^{\ominus}$'s delocalization

Inductive effect



$e^{\ominus}$ delocalization through space

Field effect



⇒ π $e^{\ominus}$ delocalization

Resonance

Mesomeric effect

Electromeric effects

Steric effect

⇒ π + σ $e^{\ominus}$ delocalization

Hyperconjugation

Inductive effect

An experimentally observable effect (on rates of reaction, etc.) of the transmission of charge through a chain of atoms by electrostatic induction. (IUPAC Goldbook)